Creating a Dataset of Autistic Individuals to Recommend Evidence-Based Practices through Generative Artificial Intelligence

Jennifer Kouo
School of Education
Johns Hopkins University
Baltimore, USA
jennifer.kouo@jhu.edu

Jinjuan Heidi Feng
Computer and Information Sciences Department
Towson University
Baltimore, USA
jfeng@towson.edu

Enric Jiao
Department of Computer Science
University of Maryland, College Park
College Park, USA
enricjiao3@gmail.com

2025 IEEE 13th International Conference on Healthcare Informatics (ICHI) | 979-8-3315-2094-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICHI64645.2025.00023

***Abstract*—Autism is the fastest-growing developmental condition, and as a spectrum disorder, individuals vary in severity in communication and behavior. The rapid development of artificial intelligence (AI) and machine learning (ML) presents opportunities to leverage such technology to support autistic individuals in multiple contexts, including education, healthcare, and disability research. However, very limited research has focused on developing AI or ML-based solutions for autistic individuals largely due to the scarcity of autism-related datasets. In order for autistic people to equally benefit from the breakthroughs in AI, the data representation gap in AI and ML must be addressed. The project proposes an innovative human expert and AI combined approach to generate synthetic data about autistic individuals. Through this approach, we are developing the first dataset of case studies of autistic individuals to effectively identify evidence-based practices (EBPs) to address the unique needs of autistic individuals. The project improves the representation and inclusion of autistic individuals in the rapidly evolving field of ML and AI through both methodological innovations (i.e., the expert/AI combined data generation approach) and practical contributions (i.e., the EBP dataset that will be made publicly accessible). The innovative data generation method and supporting materials developed in this project can be generalized to individuals with other disabilities.**

Keywords— autism, artificial intelligence, machine learning, evidence-based practice

I. Introduction and Related Research

Autism is the fastest-growing neurodevelopmental condition, with 1 in 36 children diagnosed in the United States [1]. A heterogeneous population, autistic individuals are impacted differently and with varying severity in the areas of communication, behavior, and interests [2]. Evidence-based practices (EBPs), such as those identified by the National Clearinghouse on Autism Evidence and Practice (NCAEP) and the National Autism Center (NAC) are critical for autistic individuals, their families, educators and practitioners, community members, researchers, advocacy groups, policymakers, and healthcare providers to better support autistic individuals in many contexts and situations [3, 4, 5]

Artificial intelligence in education (AIEd) has the potential to leverage novel technology to impact the learning environment and support teachers and students in teaching and learning [6]. Furthermore, the role of machine learning (ML), a subset of artificial intelligence (AI), is becoming increasingly more present and important in educational settings [6, 7]. ML is a novel technology that can rapidly analyze vast amounts of data for pattern recognition, classification, and cluster identification within a diverse population.

ML has been applied successfully in a variety of domains including self-driving cars, marketing, and early diagnosis of cancer [8]. However, applications of ML in autism-related research and services are very limited. Early efforts involving ML have primarily focused on diagnostics to reduce the time required for manual detection and diagnosis [9, 10, 11, 12]. Wireless sensors, smartphones, and cameras have been used to detect stereotypical motor movements or predict tantruming behaviors in classrooms through ML algorithms [13, 14, 15].

ML has only preliminarily been applied to detect learning activities and predict learning outcomes in education settings [16, 17]. In addition to leveraging ML in diagnosing autism, this technology has the potential to be applied to the identification of EBPs that can be used by educators, families, healthcare providers, and community members to support autistic individuals. To date, no ML-based solution exists to reduce the time and efforts needed to effectively identify EBPs to address the unique, individual needs of autistic individuals [18]. The development of such applications requires an extensive amount of data regarding the characteristics of autistic individuals, the interventions applied together with the specific context, as well as the outcome of the interventions. Such datasets about autistic individuals do not presently exist or are not available to researchers and developers due to security or privacy concerns [19]. Overall, there is a concerning lack of representation of people with disabilities, including autistic individuals, in ML and AI [20]. According to Brandsen et al. [21], existing AI language models demonstrate an alarming level of bias towards or negative associations with terms related to neurodiversity and autism. Limited and unrepresentative data availability makes it impossible to leverage, train, and evaluate ML models that support autistic individuals.

Fortunately, recent breakthroughs in pre-trained Large Language Models (LLMs) such as Generative Pre-trained Transformer (GPT) provide a highly effective approach for generating various forms of data ranging from text to videos [22, 23]. Since LLMs are trained on vast amounts of world knowledge and human examples, they have absorbed rich commonsense that makes it possible to propose reasonable plans and solve problems that require mathematical, symbolic, commonsense, and knowledge reasoning [24]. New prompt techniques such as Chain-of-Thought (CoT) have demonstrated significant enhancement in problem-solving abilities on specific tasks [25]. Reinforcement Learning through Human Feedback (RLHF) has proven to be highly

2575-2634/25/$31.00 ©2025 IEEE
DOI 10.1109/ICHI64645.2025.00023

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

effective in both training LLMs and improving the quality of LLM output [26]. Building on top of these exciting developments in LLMs, we will address the data representation gap by generating a synthetic dataset on autistic individuals through human experts and LLM combined approach.

Hinojo-Lucena et al. [30] and Hopcan et al. [6] state that a greater concerted effort to apply and research the application of AI and ML in special education is needed. Hopcan et al. [6] further urge researchers "to conduct studies on the use of AI technology for the social, emotional, and behavioral development of students with special educational needs. Also, focusing on different participant groups and designing studies to obtain large datasets is essential because the existing literature is weak in terms of researching large sample sizes". Singh et al. [7] recommend that future research explore the validation of ML to address the treatment of autism in healthcare settings. In parallel, the identification of EBPs to support autistic individuals is equally important in all contexts. This project novelly leverages ML to impact education, healthcare, and disability research by addressing the data representation gap. With the engagement of experts in the field of autism, we will use a Large Language Model to generate the first synthetic dataset of case studies of autistic individuals to effectively identify EBPs to address the unique needs of autistic individuals [18, 19].

II. Methodology

A. Autism experts

Four experts in autism and EBPs were recruited for the project. All of the experts were current or former special education teachers, had worked closely with autistic individuals, and had completed coursework on EBPs specifically for the autism population. Additional information about each of the experts is detailed in Table 1.

The phases of the project are presented in Figure 1 and detailed in the subsequent sections.

TABLE I. Information about the Autism experts

Academic background	Professional position	Years of experience	Age Range of autistic individual served	Parent/ guardian of autistic individual
M.S. in Special Education	Social Emotional Learning Teacher and Behavior Interventionist	20	3 - 21	No
M.S. in Special Education	Infants and Toddlers Teacher	9	1 - 34	No
M.S. in Special Education	Autism Content Expert	44	2 - Adult	No
M.S. in Special Education	Special Education Teacher	6	Birth - 21	Yes

B. Phase 1. Workshop with Autism Experts

Prior to the 2-hour workshop, autism experts were asked to view portions of a video entitled "State of GPT" [31] and read portions of the articles published by Hartvigsen et al. [23] and Wei et al. [25] to gain basic knowledge about LLMs and related techniques. The experts then independently identified and shared detailed, exemplary case studies accurately presenting the heterogeneity and complexity of autistic individuals. The experts were encouraged to search the literature for case studies, as well as other resources, including training modules and textbooks. The experts then reflected on and listed key components or descriptors of high-quality case studies (e.g., specific types of data that should be included to effectively describe a student with autism). Finally, the experts had the option of completing any Autism Focused Intervention Resources and Modules (AFIRM) modules to review any EBPs for autistic individuals [32].

Guided by the co-authors, the autism experts engaged in the workshop. The workshop provided an introduction to ML and LLMs, the overall project, and a brief demonstration of the capabilities of LLMs in developing case studies. A large component of the workshop involved reviewing and reflecting on the activities related to high-quality case studies with the goal of co-constructing a template or framework with prompts to develop multiple case studies of various lengths during the workshop. Discussions on the activities completed before the workshop centered around concerns regarding the availability of detailed and comprehensive case studies. The experts expressed that the case studies identified were oftentimes broad, or focused narrowly on areas of need that were being targeted with a specific intervention. One expert noted that neuropsychological testing reports may be the most comprehensive in providing a full picture of an individual with autism.

The components of the case study template developed at the end of the workshop are detailed in Table 2. The experts and the co-authors engaged in rich conversations as the template was developed. Experts sought clarification on the perspective of the case study. For example, a case study written from the perspective of an educator may include an increased amount of quantitative assessment data and be written in a manner similar to a formal, evaluative report. However, a case study from the perspective of a family member or guardian may include more anecdotal observations and therefore be more qualitative. The co-authors and experts discussed that the different perspectives would still provide valuable information when teaching the LLM model. Future expectations of the project will include both educators and family members providing information to support the automatic generation of individualized EBPs used in an educational or healthcare context. Additionally, the co-authors emphasized the importance of teaching the LLM model commonly utilized acronyms, the names of assessment tools, and clarifying the meaning or interpretation of the assessment data scores to help the LLM model make meaning of those scores and provide a comparison to a baseline of neurotypical individuals. The co-authors and experts agreed that detail in the case studies was vital to ensure that the data provided was substantive and informative. Ultimately, experts found the development of case studies to be in alignment with their day-to-day work as educators, especially with regard to developing Individualized Education Plans (IEP) and reports of student progress.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

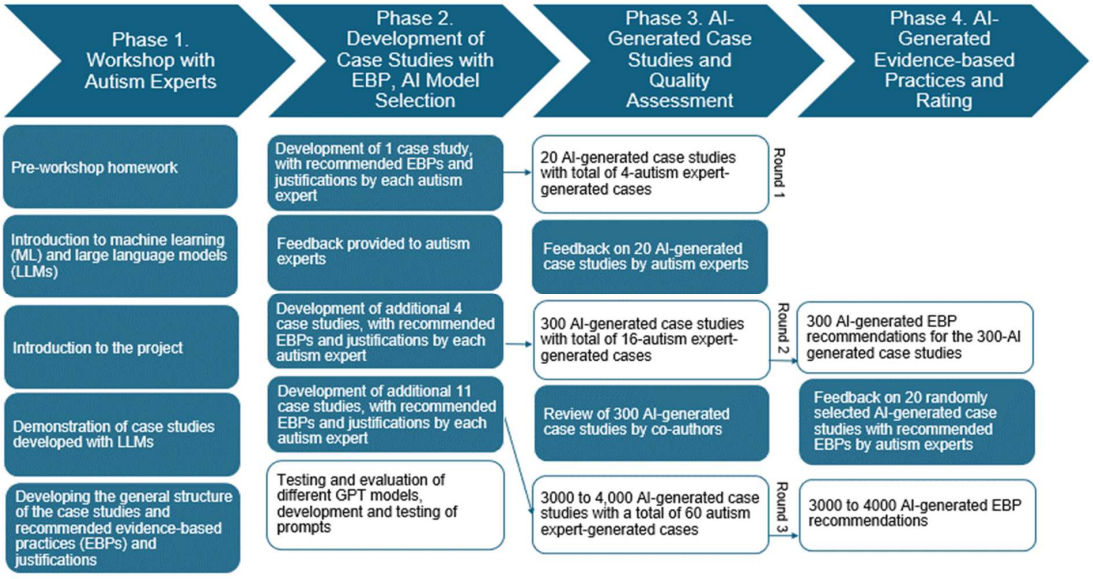


Fig. 1. Phases of the Project to Create a Dataset to Recommend Evidence-Based Practices through Generative Artificial Intelligence.

TABLE II. EXPERT CREATED TEMPLATE FOR CASE STUDY

Components of the Case Study	Additional Instructions and Considerations
• Pseudonym, Current Age, and Gender • Current School/Home Context • Brief History/Birth and Development • Comorbid Diagnosis • Areas of Strength and/or Weakness based on Assessment Data and Interpretation of the Assessment Data (*all/some of the categories below can be used with regard to need*) ○ Cognitive/Academic ○ Physical/Gross/Fine Motor Skills ○ Language Skills ○ Social Communication/Interaction ○ Behaviors/Emotional Regulation/Adaptive (stereotyped or repetitive motor movements, restricted interests, sameness, hyper- or hyporeactivity to sensory input) ○ Health	• Accurate representation of autistic individuals in the development of case studies is important. • Use headings to help organize your case study. • Case studies should be no more than 1,000 words/approximately 1 single-spaced page. • Be as concise as possible and give priority to Areas of Strength and/or Weakness based on Assessment Data and Interpretation of the Assessment Data. • Other Considerations: ○ Draw upon experts' knowledge of autistic students ○ Consider the elements of the Individualized Education Plan (IEP) ○ Consider the broad spectrum of autism ○ Consider various ages ○ Think about how students have developed across a school year or over multiple years ○ Develop case studies that are an amalgamation of students ○ Ensure anonymity to protect the privacy and identity of students

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

TABLE III. EXPERT CREATED TEMPLATE FOR EVIDENCE-BASED PRACTICE RECOMMENDATIONS

Components of the Template	Additional Instructions and Considerations
• Evidence-based Practice 1 - Restate evidence from the case study (e.g., needs, strengths, interests, context). • Evidence-based Practice 2 - Restate evidence from the case study (e.g., needs, strengths, interests, context). • Evidence-based Practice 3 - Restate evidence from the case study (e.g., needs, strengths, interests, context).	• All evidence-based practices identified and accompanying justifications should be no more than 300 words in total.

As part of the project aims, experts were asked to identify three priority EBPs that would be responsive to the needs identified in each case study they would be developing. Additionally, experts were required to provide a rationale or justification for selecting each EBP by referring directly back to the case study information. Providing a clear justification would further teach the ML model to connect the case study content with the selection of specific EBPs, as suggested by the 'chain-of-thought' prompting technique [25]. Table 3 contains the template developed during the workshop

C. Phase 2. Development of Case Studies with Justified Evidence-based Practices

Following the workshop, each expert was asked to develop one case study, including recommended EBPs and their justification. The co-authors reviewed the four generated case studies and provided feedback for the following case studies that would be developed. The feedback included the removal of identifying information including a city and state. Experts were asked to provide the full term of an acronym when first mentioned, followed by the acronym itself written in capital letters and enclosed by parentheses. References to assessment tools in the case studies required at least one sentence describing the tool. Concerning recommending EBPs, the co-authors asked experts to refer to the EBPs and definitions indicated in Steinbrenner et al. [5] when recommending EBP(s). After identifying the EBP from that table, experts were encouraged to provide more detail about the EBP, if applicable. For example, the recommendation of the EBP prompting could contain additional detail about the type, including gestural, visual, or physical prompting. To make it obvious to the LLM model, the authors asked experts to cite or quote directly from the case study what the recommended EBP would address. This change was expected to help make a clearer connection between the case study and the EBP being recommended. Applying the additional feedback, the experts each generated an additional four case studies with accompanying EBP recommendations and their justification.

Since the proposed approach is highly innovative, there is no previous work that may provide guidance on the process or insight into the possible quality of the LLM output. An iterative approach to data generation was undertaken to ensure that the AI-generated data was continually evaluated by the co-authors and autism experts to inform the next stage of data generation. Within Phase 2 and as presented in Figure 1, the autism experts generated additional case studies and recommended EBPs in increments. The increments were followed by the development of AI-generated case studies, which are organized by rounds and further described in the next section on Phase 3.

D. Phase 3

Round 1. Utilizing the initial four case studies developed by the autism experts in Phase 2, a total of 20 case studies were generated using the GPT-3.5 Turbo model with the following prompt:

> "You are an ASD researcher. For the purpose of ASD research, we would like you to generate several young student or patient examples based on the following examples. 1. Please vary the demographics such as name, gender and age. 2. Please do not repeat the patient background. 3. Please generate information that is often observed in ASD students or patients. 4. Please keep each case to be between 500 to 1000 tokens."

The number of cases generated during each cycle was set at two or three. The length of the generated case was set to 500 to 1000 tokens (approximately 375 to 750 words). These ranges were selected considering the input and output limits of the GPT-3.5 Turbo model.

After the 20 cases were generated the experts were asked to read all 20 case studies and to consider whether or not the cases were diverse. The experts provided their responses in a shared document. Then each expert was assigned five of the 20 case studies to read thoroughly. Experts provided a simple rating (Yes, No, Maybe) for each case study for the following questions: (1) Is the case realistic (i.e., does not contain contradictory information)?; (2) Is the case representative of an autistic individual?; and (3) Does the case contain sufficient information to provide an EBP recommendation? The experts were encouraged to provide a rationale for their ratings. The expert rating suggested that the overall quality of the 20 cases was satisfactory.

Round 2. After the first round of quality checks in Round 1, the autism experts each developed four additional case studies. The GPT-3.5 Turbo model was used to generate 300 cases using the 16 total cases created by the autism experts using the same prompt and length ranges used in Round 1. The co-authors reviewed the 300 cases and agreed that the general quality of the cases was satisfactory.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

TABLE IV. SUMMARY OF EXPERT RATINGS FOR AI GENERATED CASE STUDIES

	AI generated case is realistic	AI generated case is representative	AI generated case contains sufficient information for EBP recommendation
Both experts agree	13 (65%)	17 (85%)	19 (95%)
One expert agree	6 (30%)	2 (10%)	1 (5%)
Both experts Disagree	1 (5%)	1 (5%)	0

TABLE V. SUMMARY OF EXPERT RATINGS FOR EVIDENCE-BASED RECOMMENDATIONS

	1st recommendation	1st justification	2nd recommendation	2nd justification	3rd recommendation	3rd justification
Both experts agree	15 (75%)	17 (85%)	14 (70%)	16 (80%)	16 (80%)	17 (85%)
One expert agree	5 (25%)	3 (15%)	5 (25%)	4 (20%)	3 (15%)	3 (15%)
Both experts Disagree	0	0	1 (5%)	0	1 (5%)	0

Round 3. In the final round, using all 60 cases created by the experts as examples, the GPT-3.5 turbo model will be utilized to generate approximately 3,000 to 4000 cases.

E. Phase 4 AI-Generated Evidence-based Practices and Rating

After the quality check conducted by the co-authors on the 300 AI-generated cases, the GPT-3.5 Turbo model was used to generate EBP recommendations for the 300 AI-generated cases. The following prompt was used as instructions for the model:

> "You are an ASD researcher. For the purpose of ASD research, we would like you to generate three evidence-based practices (EBP) for a scenario. You will be given a few scenarios with expert-generated EBP, and several other scenarios without EBP. Your role is to create three EBPs appropriate to the scenario."

After the EBPs were generated for the 300 cases, 20 cases together with the EBP recommendation and justifications were randomly selected and rated by the autism experts. Each autism expert was assigned 10 case studies to review independently, with each case study being reviewed by 2 experts to explore the consistency of ratings. The experts were asked to respond to the following questions with a simple rating (Yes, No) for each AI-generated case study: (1) Is the case realistic (i.e., does not contain contradictory information)?; (2) Is the case representative of an individual with autism?; and (3) Does the case contain sufficient information to provide an EBP recommendation?. Experts then provided a rationale to support the ratings. For each AI-recommended EBP, autism experts were asked to rate (Yes, No) whether the recommendation was appropriate based on the case study information, and if the justification was accurate. After experts reviewed all 10 assigned case studies, they were asked to describe whether all 10 cases reviewed were diverse.

III. RESULTS

The proposed data generation approach requires close collaboration between AI and experts with experience working with autistic students in special education settings and with knowledge of EBPs responsive to the needs of autistic individuals. Case studies and recommended EBPs were generated by autism experts to inform the development of AI-generated data.

A. Sample expert generated case with EBP

Table 6 in the Appendix presents an example of a case study developed by an autism expert during Phase 2 of the project.

B. Sample AI generated case with EBP

Based on 16 expert created case studies from Phase 2, 300 case studies with EBPs were generated by the GPT 3.5 Turbo model. An example is presented in table 7 in the Appendix.

C. Quality assessment: Case studies

In Phase 4, 20 AI-generated case studies together with the corresponding EBP recommendations and justifications were randomly selected among the 300 cases and rated by the autism experts. In order to determine interrater reliability, each case study was reviewed by two experts.

Table 4 provides a summary of the expert ratings for the 20 randomly selected AI generated case studies. The experts were asked to provide three ratings for each case study: whether it is realistic; whether it is representative; and whether it provided sufficient information for EBP recommendation. There was a 65% agreement that the case studies were realistic. The rationale for rating the case study as unrealistic

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

centers upon the educational setting identified and contradictory assessment and descriptive data. Regarding the setting, experts indicated that the AI-generated cases identified a general education setting when areas of need in the case study (e.g., behaviors during transitioning, support with toileting) would likely require a more restrictive setting or a setting with additional adult support through a paraprofessional. Another expert noted that an AI-generated case study referenced the use of a "versatile special education service," which is not a known school setting or learning environment. Autism experts also identified contradictions as it related to assessment data reported in the case studies. Despite the AI-generated case study indicating the age range for an assessment, an expert noted, "This case uses an assessment that is not appropriate for the age of the individual. It states that Tara is 14 years old and then describes the use of an assessment for children birth to 5 years of age." Another autism expert noted that in another case study, an average reading score was reported but later indicated a large gap in terms of age and grade level reading ability. There was a 85% agreement that the case studies were representative of an autistic individual. Experts reported that the AI-generated case studies did not include a definitive description of an autistic individual and that the case study could also be representative of a neurotypical child or a child with delays only in communication. There was a 95% agreement that the AI-generated case studies contained sufficient information to recommend EBPs, with one case study being identified as lacking information.

D. Quality assessment: EBP recommendations

Experts also rated each of the three AI-generated EBPs as well as their justifications for the 20 cases (Table 5). For each EBP, experts were asked: whether the EBP recommendation was appropriate; and whether the justification for the EBP was accurate.

For the first AI-recommended EBP, there was 75% agreement on its appropriateness. For the second and third EBP recommendations, there was 70% and 80% agreement respectively. Autism experts provided critiques about the appropriate labeling of EBPs. For example, experts suggested that visual supports should first be identified before specifically recommending visual schedules or identifying augmentative and alternative communication (AAC) prior to recommending Picture Exchange Communication Systems (PECS). Experts also noted for another case study the appropriateness of the social stories if the case study indicated the individual struggled with attending to stories and had delays in receptive language.

When reviewing the accuracy of the justification generated by AI, there was 85% agreement for the first EBP recommended, followed by 80% and 85% agreement for the second and third EBP recommendations respectively. Autism experts noted that the justifications needed to go beyond providing a general definition of the EBP and identify specifically how the EBP would address the needs of the individual described in the case study. Experts identified a lack of specific justifications for Applied Behavior Analysis (ABA), peer-mediated instruction and intervention (PMII), and visual supports. For example, an autism expert provided the following recommendation to improve the AI-generated justifications: "Visual Supports are an evidence-based practice that utilizes pictures, objects or images to support an individual in understanding and engaging with their environment. Since Clara struggles with anxiety around changes in her routine, a visual schedule could be beneficial in preparing her for changes that may occur in her day, thus lessening her anxiety." Another recommendation to improve justifications included, "ABA is an evidence-based practice that uses the principles of learning to increase or decrease target behaviors or skills. This strategy could be beneficial in addressing Oliver's aggressions and self-injurious behaviors. Methods used with ABA to address these behaviors may include reinforcing the absence of these behaviors, teaching and reinforcing behaviors that interfere with the target behavior, and teaching and reinforcing alternate behaviors that are more appropriate based on the function of the behavior (e.g. asking for attention)." Autism experts also noted that there were a few cases of inaccurate justification, where information in the justification did not reflect the case study information.

After each autism expert reviewed their 10 assigned case studies, experts were asked to describe whether or not all 10 cases reviewed were diverse. Experts indicated that the case studies generated by AI were diverse with regard to ages, abilities, and areas of need. However, it was noted that the cases reviewed did contain more examples of individuals with higher-intensity needs and that there was limited cultural diversity among the cases.

IV. Discussion

The reported work is the first proof-of-concept project that uses the recent generative AI technology to generate synthetic data about autistic individuals and make recommendations for EBPs. The preliminary result is promising, with highly positive expert agreements for both case study generation and EBP recommendations and justifications.

A. Significance

The project novelly applies generative AI technology to autism data generation and the identification of EBPs that can be used to support professionals in various fields such as education and healthcare. It will lead to two major outcomes that will immediately impact both the fields of generative AI and autism research: (1) the first publicly available datasets regarding EBPs for autistic individuals that all researchers, practitioners, and stakeholders can use for their own research and practice, such as curriculum development, training, and the adoption of LLMs related applications; and (2) a new method for creating substantial amount of synthetic data regarding autistic individuals through the collaboration of domain experts and AI/LLMs. The new LLM/expert-combined data generation method will have a long-lasting impact on the autism community regarding data representation and the adoption of AI/LLMs in the autism domain. Researchers and practitioners can use this method to generate data at comparatively low cost and develop specialized LLM models and assistants for healthcare, education, workplace, and other contexts. The LLM/expert combined data generation method can also be generalized beyond the autism community to other types of disabilities.

The practical impacts of the project on autistic individuals span multiple disciplines. In the healthcare domain. The EBP recommendation model may be leveraged to support healthcare professionals in providing patient- and family-centered care by identifying effective and responsive EBPs during medical care visits. The findings from the project may also be translational for the education and workplace domain.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

With the nationwide shortage of educational evaluators, special and general education teachers, and a current process that is time-intensive, the development of a tool to rapidly and effectively identify EBPs for autistic students is a game changer. Within the workplace, managers and colleagues may utilize the outcomes of this project, in conjunction with additional education and training, to better support autistic employees. In sum, the project and future work generated from it will be able to improve the quality of life of autistic individuals, increase the understanding and support of this disability population, and improve inclusion and acceptance across different contexts.

B. Limitations

The project and results are in its initial stages and include limitations that are expected to be addressed in future research. The reported work only engaged 4 autism experts for training data generation and the validation of AI-generated cases and EBPs. The co-authors recognize that a much larger number of diverse stakeholders must be engaged in the training and validation process, including autism researchers, educators, family members, caretakers, and autistic individuals themselves, to reduce potential bias in the autism expert-created sample data.

Additionally, Recent research suggested that, without enough real data in each generation of the autophagous loop, the quality and diversity of the generated data will gradually decrease [33]. We are planning to control this risk through two approaches: 1) collecting real-life data about autistic individuals through an online platform to supplement the expert-generated real data as examples to teach the LLMs in data generation, and 2) using various LLMs, such as Llama, GPT, and BERT, to generate data. In this way, the amount of data generated by each model is comparatively low, avoiding significant deterioration in data quality. This approach will also help control the potential bias contributed by each of the LLMs.

Thirdly, only a small number of AI generated cases and EBPs have been evaluated by human experts. Human evaluation of a much larger number of AI-generated cases is necessary to validate the effectiveness of the data generation and EBP recommendation functions.

C. Future directions

Generative AI technology is rapidly evolving and research will need to evolve accordingly. The proposed data generation approach will need to be refined to reflect the continual development of generative AI. Future directions will develop the first tailored approach to fine-tuning transformer models for identifying EBPs for autistic individuals using a limited amount of human-labeled data and incomplete data through active learning. In order to enrich the training data, more expansive data from stakeholders will need to be gathered, including case studies generated by educators, families, and autistic individuals [21]. Additionally, field studies to assess the recommendation model in both education and healthcare contexts are needed. Data collected from the field studies will be used as additional training data for model fine-tuning to further improve the accuracy of the recommendation model.

V. CONCLUSIONS

Recent breakthroughs in ML and AI signal timeliness for the project. The growing presence of AI highlights the importance of leveraging this powerful tool to positively impact the lives of autistic individuals, their families, and all who work closely with this population. This project proposed and demonstrated an innovative human expert and AI combined approach to generate synthetic data about autistic individuals to support the identification of responsive EBPs. The preliminary results suggest that data generated had promising quality. Through this approach, we are developing the first dataset of case studies of autistic individuals to effectively identify EBPs to address the unique needs of autistic individuals. This project has notable impacts in both the education and healthcare context. Accurate and rapid identification of EBPs can lead to targeted intervention implementation to support skill development in educational settings and provide more patient- and family-centered care during medical interactions.

ACKNOWLEDGMENT

This work was supported by the Thomas Wilson Sanitarium for Children of Baltimore City Foundation and the School of Emerging Technologies of Towson University and the Faculty Development and Research Committee at Towson University. We also would like to thank all the Autism experts who contributed to the work.

REFERENCES

[1] Centers for Disease Control. (2023, April 4). Data & Statistics on Autism Spectrum Disorder. https://www.cdc.gov/ncbddd/autism/data.html

[2] American Psychiatric Association. (2013). Diagnostic and statistical manual of mental disorders (5th ed.). Washington, D.C.: American Psychiatric Association.

[3] National Autism Center. (2015). Evidence-based practice and autism in the schools (2nd ed.): An educator's guide to providing appropriate interventions to students with autism spectrum disorder. Randolph, MA: Author.

[4] Steinbrenner, J., Sam, A., Chin, J., Morgan, W., & AFIRM for Paras Team. (2019). Introduction to ASD. FPG Child Development Institute, University of North Carolina. Retrieved from https://afirm.fpg.unc.edu/introduction-asd

[5] Steinbrenner, J. R., Hume, K., Odom, S. L., Morin, K. L., Nowell, S. W., Tomaszewski, B., Szendrey, S., McIntyre, N. S., Yücesoy-Özkan, S., & Savage, M. N. (2020). Evidence-based practices for children, youth, and young adults with Autism. The University of North Carolina at Chapel Hill, Frank Porter Graham Child Development Institute, National Clearinghouse on Autism Evidence and Practice Review Team.

[6] Hopcan, S., Polat, E., Ozturk, M. E., & Ozturk, L. (2023). Artificial intelligence in special education: A systematic review. Interactive Learning Environments, 31(10), 7335–7353. https://doi.org/10.1080/10494820.2022.2067186

[7] Singh, A., Farooqui, Z., Sattler, B., Li, E., Nerkar, S., Helde, M., & Usua, U. (2023). Machine learning in autism spectrum disorder diagnosis and treatment: techniques and applications. In Suri, J. S., & S. El-Baz, A. (Eds.), Neural engineering techniques for autism spectrum disorder. Volume 2, Diagnosis and clinical analysis (pp. 173-193). Academic Press. https://doi.org/10.1016/B978-0-12-824421-0.00016-3

[8] Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future - big data, machine learning, and clinical medicine. The New England Journal of Medicine, 375(13), 1216–1219. https://doi.org/10.1056/NEJMp1606181

[9] Abbas, H., Garberson, F., Glover, E., & Wall, D. P. (2018). MACHINE LEARNING approach for early detection of autism by combining questionnaire and home video screening. Journal of the American Medical Informatics Association, 25(8), 1000-1007.

[10] Noorbakhsh-Sabet, N., Zand, R., Zhang, Y., & Abedi, V. (2019). Artificial Intelligence Transforms the Future of Health Care. The American Journal of Medicine, 132(7), 795–801. https://doi.org/10.1016/j.amjmed.2019.01.017

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

[11] Thabtah, F., & Peebles, D. (2020). A new machine learning model based on induction of rules for autism detection. Health Informatics Journal, 26(1), 264–286. https://doi.org/10.1177/1460458218824711

[12] Uddin, Md. Z., Shahriar, Md. A., Mahamood, Md. N., Alnajjar, F., Pramanik, Md. I., & Ahad, M. A. R. (2024). Deep learning with image-based autism spectrum disorder analysis: A systematic review. Engineering Applications of Artificial Intelligence, 127, 107185-. https://doi.org/10.1016/j.engappai.2023.107185

[13] Albinali, F., Goodwin, M. S., & Intille, S. (2012). Detecting stereotypical motor movements in the classroom using accelerometry and pattern recognition algorithms. Pervasive and Mobile Computing, 8(1), 103-114.

[14] Patnam, V. S. P., George, F. T., George, K., & Verma, A. (2017). Deep Learning Based Recognition of Meltdown in Autistic Kids. The Institute of Electrical and Electronics Engineers, Inc. (IEEE) Conference Proceedings, 391-396.

[15] Rad, N. M., Kia, S. M., Zarbo, C., van Laarhoven, T., Jurman, G., Venuti, P. & Furlanello, C. (2018). Deep learning for automatic stereotypical motor movement detection using wearable sensors in autism spectrum disorders. Signal Processing, 144, 180-191.

[16] Linstead, E., German, R., Dixon, D., Granpeesheh, D., Novack, M., & Powell, A. (2015). An Application of Neural Networks to Predicting Mastery of Learning Outcomes in the Treatment of Autism Spectrum Disorder. The Institute of Electrical and Electronics Engineers, Inc. (IEEE) Conference Proceedings, 414-418.

[17] Zheng, H., Mahapasuthanon, P., Chen, Y., Rangwala, H., Evmenova, A. S., & Genaro Motti, V. (2021). WLA4ND: A Wearable Dataset of Learning Activities for Young Adults with Neurodiversity to Provide Support in Education. The 23rd International ACM SIGACCESS Conference on Computers and Accessibility, 1-15.

[18] Hyde, K. K., Novack, M. N., LaHaye, N., Parlett-Pelleriti, C., Anden, R., Dixon, D. R., & Linstead, E. (2019). Applications of supervised machine learning in autism spectrum disorder research: A review. Review Journal of Autism & Developmental Disorders, 6(2), 128.

[19] Kamikubo, R., Wang, L., Marte, C., Mahmood, A., & Kacorri, H. (2022, October). Data Representativeness in Accessibility Datasets: A Meta-Analysis. Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility, 1-15.

[20] Kacorri, H., Dwivedi, U., Amancherla, S., Jha, M., & Chanduka, R. (2020, October). IncluSet: A data surfacing repository for accessibility datasets. The 22nd International ACM SIGACCESS Conference on Computers and Accessibility, 1-4.

[21] Brandsen, S., Chandrasekhar, T., Franz, L., Grapel, J., Dawson, G., & Carlson, D. (2024). Prevalence of bias against neurodivergence - related terms in artificial intelligence language models. Autism Research, 17(2), 234‑248. https://doi.org/10.1002/aur.3094

[22] Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). Improving language understanding by generative pre-training.

[23] Hartvigsen, T., Gabriel, S., Palangi, H., Sap, M., Ray, D., & Kamar, E. (2022). ToxiGen: A Large-Scale Machine-Generated Dataset for Adversarial and Implicit Hate Speech Detection. arXiv.Org. https://doi.org/10.48550/arXiv.2203.09509

[24] Huang, W., Abbeel, P., Pathak, D., & Mordatch, I. (2022, June). Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In International Conference on Machine Learning (pp. 9118-9147). PMLR.

[25] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2023). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. arXiv.Org. https://doi.org/10.48550/arxiv.2201.11903

[26] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). Training language models to follow instructions with human feedback. Advances in Neural Information Processing Systems, 35, 27730-27744.

[27] Gernsbacher, M. A. (2017). Editorial perspective: The use of person first language in scholarly writing may accentuate stigma. Journal of Child Psychology and Psychiatry, 58(7), 859–861. https://doi.org/10.1111/jcpp.12706

[28] Robison, J. E. (2019). Talking about autism—Thoughts for researchers. Autism Research, 12(7), 1004–1006. https://doi.org/10.1002/aur.2119

[29] Shakes, P., & Cashin, A. (2019). Identifying language for people on the autism spectrum: A scoping review. Issues in Mental Health Nursing, 40(4), 317–325. https://doi.org/10.1080/01612840.2018.1522400

[30] Hinojo-Lucena, F. J., Aznar-Díaz, I., Cáceres-Reche, M. P., & Romero-Rodríguez, J. M. (2019). Artificial intelligence in higher education: A bibliometric study on its impact in the scientific literature. Education Sciences, 9(1), 1–9. https://doi.org/10.3390/educsci9010051

[31] Microsoft Developer. (2023, May 25). State of GPT [Video] YouTube. https://www.youtube.com/watch?v=bZQun8Y4L2A.

[32] AFIRM Team. (2019). Components of the Autism Focused Intervention Resources & Modules (AFIRM). Chapel Hill, NC: National Professional Development Center on Autism Spectrum Disorder, Frank Porter Graham Child Development Center, University of North Carolina. Retrieved from https://afirm.fpg.unc.edu/afirmmodules

[33] Alemohammad, S., Casco-Rodriguez, J., Luzi, L., Humayun, A. I., Babaei, H., LeJeune, D., ... & Baraniuk, R. G. (2023). Self-consuming generative models go mad. arXiv preprint arXiv:2307.01850.

Appendix.

A.1 Table VI. Expert created case study with Evidence-based recommendation

Case Study	Evidence-based Practices and Justification
Name: Trent Current Age: 10 Gender: Male Current School/Home Context: Trent attends a public separate day school which serves students ages 3-21 with multiple developmental disabilities including autism. His class consists of 7 students, including Trent, 1 special education teacher, 1 paraeducator, and 3 additional adult assistants. Schoolwide supports include the use of evidence-based practices in a highly structured environment, high staff to student ratio, integrated related services support for SLP, OT, PT, Social Work, SEL, and Nursing. Trent lives at home with his parents and older sister who receives special education services through an Individualized Education Plan. Brief History/Birth and Development: Trent was born at 39 weeks gestation weighing 6 pounds 11 ounces via cesarean section. The pregnancy and delivery	Evidence-based Practice 1 – Prompting (PP) – Prompting refers to verbal, gestural, modeling, or physical assistance given to learners to support them in acquiring or engaging in a targeted behavior or skill. Because Trent's difficulty with communication causes frustration which sometimes leads to self-injurious behavior defined as hitting himself in the head with a closed fist, prompting effective and safe

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

were unremarkable. He passed the newborn hearing screening and was released to home following a short hospital stay. Trent was slow to acquire motor and language milestones and by 18 months of age, was referred to an Infant and Toddler Program for developmental evaluations. Specialized instruction designed to improve Trent's ability to engage with others, play with a wider variety of toys, ambulate independently, increase his vocalizations, and develop self-regulation skills was initiated at this time. By the age of 4, Trent was diagnosed with Autism Spectrum Disorder Level 2.

Comorbid Diagnosis: Type 1 Diabetes

Areas of Strength and/or Weakness based on Assessment Data and Interpretation of the Assessment Data

Cognitive/Academic: Trent completed the nonverbal portion of the Stanford Binet Intelligence Scale Fifth Edition (SB-V), which is an intelligence test for individuals between 2 years and 85 years of age. Trent's SB-V nonverbal question responses resulted in a nonverbal intelligence quotient (NVIQ) of 42, which falls within the moderately impaired range on a scale from very gifted to moderately impaired. Trent is currently performing below kindergarten level in reading and below grade 1 in math.

Physical/Gross/Fine Motor Skills: Trent is left hand dominant, and he is able to consistently use an age-appropriate grasp on his writing tools. When copying from a visual model to write his first name, Trent is able to copy "T" and "r" consistently; however, for the remaining letters "e" "n" and "t" he tends to scribble when writing them. When Trent is provided proprioceptive/tactile input (therapist hand on end of pencil or partially on his hand for control), he is able to form the letters legibly. Trent can self-feed using a spoon and/or a fork. He is independent with scooping food and bringing it to his mouth. He can drink from a water bottle, cup with a straw and an open cup independently. For toileting, Trent is independent with pulling up and down his pants for clothing management. He occasionally receives assistance for thorough hygiene and is provided verbal reminders to use the restroom throughout the day to decrease chances of accidents. Trent demonstrates school-based mobility skills that serve him well in the supervised environment at school. While he demonstrates gait deviations related to lower leg out-toeing, he is capable of independent walking and running, stair negotiation, and playground skills. His family reports that Trent is able to climb cargo nets and rock walls, and that they enjoy hiking together.

Language Skills: According to the Peabody Picture Vocabulary Test Fourth Edition (PPVT-4), which is a single word receptive vocabulary test, Trent demonstrated scattered skills up to age 4. Overall, Trent demonstrates comprehension of some basic concepts with limitations. For example: he can follow one-step directions with 100% accuracy independently such as sit, open mouth, hands up, hands down, stand, touch, clap, and put in. He can follow two-step directions with 60% accuracy with verbal and model prompting. Trent presents with severe expressive language delays characterized by difficulty expressing his wants and needs consistently. He typically uses gestures, facial expressions, and vocalizations to communicate. Trent also often uses basic signs to communicate (e.g., more, finished, book, music, etc.) and is able to use a larger vocabulary via sign-language given prompts. He has been trialing a dynamic display device to express himself with some independence for wants (e.g., swing, water).

Social Communication/Interaction: Trent enjoys playing with cars, ramps, and toys that light up, spin, move, or make noise. He also likes to engage in activities involving music and movement. He does not interact directly with peers but will play alongside them and will share with them by putting an

forms of communication may help improve his communication skills and decrease the frustration that leads to self-injurious behavior. Staff can prompt the use of Trent's communication device or sign language by using verbal reminders, gestures, modeling, or physical assistance to teach and remind him of means to getting his needs met. Prompting would be even more effective if it is combined with reinforcement (see evidence-based practice 2). When staff prompt safe communication and Trent follows the prompt by performing the task, staff can reinforce his behavior with high quality praise and rewards.

Evidence-based Practice 2 – Reinforcement (R) – Reinforcement is the application of a consequence following a learner's use of a response or skills that increases the likelihood that the learner will use the response/skills in the future. Because Trent's difficulty with communication causes frustration which sometimes leads to self-injurious behavior defined as hitting himself in the head with a closed fist, staff can use reinforcement techniques to encourage his development of safe and effective communication skills which will decrease his frustration when attempting to get his needs met. Reinforcement would be even more effective if it is combined with prompting (see evidence-based practice 1). When staff prompt safe communication and Trent follows the prompt by performing the task, staff can reinforce his behavior with high quality praise and rewards.

Evidence-based Practice 3 - Social Skills Training (SST) - SST is effective to teach individuals with ASD foundational social skills such as initiating interactions and understanding and regulating

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

object in front of them and then looking at the peer and the object. On the playground, Trent will run around peers while they play “tag” or “hide-and-seek” but will not directly engage in the games with them. Behaviors/Emotional Regulation/Adaptive: Trent is able to follow his routine throughout the school day with verbal and visual supports and occasional calm redirection. Sometimes, however, his difficulty with communication causes frustration which may lead to self-injurious behavior defined as hitting himself in the head with a closed fist. Staff respond by blocking Trent’s aggression and reminding him to use his signs or his communication device to express his wants or needs. Health: No health concerns are reported outside of Trent’s Type 1 Diabetes.	emotions. Because Trent does not interact directly with peers, SST may be an effective tool to support his ability to engage in age-appropriate play with others. Additionally, it may support Trent’s emotional self-regulation skills by teaching him safe and effective ways to express frustration.

A.2 Table VII. GPT created case study with Evidence-based recommendation

Case Study	Evidence-based Practices and Justification
Background Information: Amelia is a 28 month old female who lives with her single mom and 10 year old sister. Amelia was born at full term with a birth weight of 7 lbs 4 oz. For her, pregnancy was eventless and she passed her audiological assessment at birth, with no concerns regarding her vision emerging since then. Lately, Amelia has been suffering from recurrent ear infections and thereby an audiological assessment is advised to eliminate any eventual hearing issues. Amelia's pediatrician referred her to Infants and Toddlers due to concerns about her behaviors that fall outside of what's expected for her age. Developmental Assessment Data: The Developmental Assessment of Young Children Second Edition (DAYC-2) was administered to gather information about Amelia’s current skill level. The DAYC-2 is a normed based assessment for children ages birth to 5 years. For a 28-month-old child like Amelia, an age-equivalent score of 21 months or below is classified as a 25% delay. Amelia's cognition, communication, adaptive behavior, physical development, and social-emotional development are probed in five main areas. Overall, Amelia's results demonstrate delays in the areas of cognition, expressive and receptive language skills - approximated to the level of a 20 and 18 month-old respectively, social and emotional skills which are suggestive of 18 months, and fine motor skills which appear to be on par with an average 20-month-old. Development of Positive Social-Emotional Skills: Amelia shows affection towards her mom and sister by giving spontaneous hugs. She is also capable of showing empathy; for instance, when her sister is upset Amelia would go to her, hug, and pat her back. Amelia’s interactions with her peers are minimum, as she would quickly turn away or ignore their attempts to engage her in play. When upset, Amelia tends to withdraw herself and avoid eye contact until she feels better. Acquisition and Use of Knowledge and Skills: Amelia likes to stack her blocks and could count up to three. She has been observed to line up her dolls repeatedly and display distress when the order is changed. She finds it difficult to engage with new toys and show minimal interest in books, often pushing them away when handed to her. While Amelia can mimic animal sounds, she uses only a few meaningful words such as 'mama' and 'no'. Her use of gestures like pointing, shaking her head for yes or no, are also	Evidence-based Practice 1 - Functional Communication Training (FCT): Due to Amelia's difficulty in using meaningful words and minimal use of gestures to communicate, FCT can be useful to assist her. FCT is a method of intervention that teaches individuals to use more adaptive and effective forms of communication than challenging behaviors. This will promote her language skills, positively affect her behavior (e.g., reduce tantrums due to frustration from an inability to express herself), and improve her interaction with peers. Evidence-based Practice 2 - Social Skills Training (SST): SST could be beneficial for Amelia to encourage her interaction with peers as she currently has minimal engagement with them. Through supervised play and structured activities designed to enhance socialization, she can learn to contribute to the group, recognize social cues, and relate to her peers. This can provide her an opportunity to develop critical social-emotional skills. Evidence-based Practice 3 - Visual Supports (VS): Given Amelia's limited verbal language skills and her resistance towards books, visual aids might be an effective tool for her. VS includes using pictures, visual schedules or social stories to

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.

minimal. Ability to Take Appropriate Action to Meet Needs: Amelia can feed herself using a spoon, although she often gets overwhelmed with larger pieces of food. She can take off her clothes unaided and tries to put on her shoes, though struggles with it. She also shows signs indicating that she might need a diaper change namely, looking uncomfortable and touching her diaper area. Amelia drinks independently from a sippy cup and keeps trying to sip from a regular cup. She enjoys bath time and attempts to wash herself, but needs direct supervision as she tends to splash around a lot. When it comes to sleep, Amelia prefers to have her bedtime story before she sleeps, though picks at the pages rather than engages with the pictures.	communicate and clarify expectations. This can help her to understand new activities, establish routines, transition between activities smoothly, and reduce her anxiety related to unpredictability. The visual support can also indirectly promote language acquisition by associating words with pictures and real-world objects.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 10,2025 at 15:37:41 UTC from IEEE Xplore. Restrictions apply.